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Big Data

Sentiment in Earnings Calls and Realized Volatility

FinBERT Approach on S&P 500 (2016–2025)

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Abstract

We investigate whether the cautious tone of S&P 500 earnings-call transcripts predicts short-term realized equity volatility beyond standard financial controls. Starting from 113,990 deduplicated transcripts (2016–2025), we build a full NLP pipeline: Koyfin scraping, knowledge-distillation labeling combining *financial_phrasebank* expert annotations and LLM-generated labels, FinBERT fine-tuning, tone-score aggregation, and panel OLS with sector, quarter, and crisis fixed effects. Across 10,226 firm-quarter observations, financial controls explain 44.5% of post-call realized volatility. Adding `cautious_share` leaves R^2 unchanged ($\hat{\beta}_3 = 0.0007$; $p = 0.945$). We interpret this as a methodologically informative null result: cautious language in scripted prepared remarks appears to be efficiently priced, leaving no detectable residual signal in ex-post realized volatility.

1 Introduction

Earnings conference calls are among the most information-dense corporate events in financial markets. Beyond hard numbers, the language employed by executives — the degree of caution in forward-looking statements, the evasiveness in responses to analyst questions — may convey incremental information about firm prospects. A growing literature exploits computational text analysis to extract such signals (Price et al., 2012; Huang et al., 2014; Hassan et al., 2019).

Research question: *Does the cautious tone of S&P 500 earnings-call transcripts predict short-term realized equity volatility, after controlling for historical volatility and the VIX?*

Prepared remarks are carefully drafted in advance; the Q&A session, by contrast, exposes management to unscripted follow-up that may reveal information through hesitation or defensiveness. If cautious language conveys genuine uncertainty, markets should respond with heightened volatility. If the signal is already embedded in price dynamics, a null result is equally informative and, as Section 4 shows, that is precisely what we find.

Our contribution is threefold: (i) a large corpus of 113,990 unique transcripts from Koyfin covering S&P 500 constituents over 2016–2025; (ii) a complete NLP pipeline combining knowledge-distillation labeling with FinBERT fine-tuning; and (iii) panel OLS regressions linking tone scores to post-call realized volatility with rigorous econometric controls.

2 Data and Pipeline

Transcript corpus. Earnings-call transcripts were scraped from Koyfin via a Selenium-based pipeline, yielding 114,466 files before deduplication and **113,990 unique transcripts** after. Restricting to S&P 500 index members via fuzzy ticker matching (`rapidfuzz`) produces **14,963 filtered transcripts** across **488 unique firms** and all 11 GICS sectors (Financials: 2,403; Industrials: 2,306; IT: 2,074; Health Care: 1,764). Full pipeline counts are reported in Table A1 (Appendix).

Market data. Daily prices from Yahoo Finance (`yfinance`) are used to compute two annualized volatility measures: `rv_post` (standard deviation of daily log-returns over the 20 trading

days *following* each call, $\times\sqrt{252}$) and `rv_hist` (same measure over the 60 days *preceding* the call). The VIX on the call date proxies aggregate uncertainty. The financial panel contains **14,847 observations** across **486 tickers** (January 2016–November 2025). After merging with FinBERT features, the regression sample comprises **10,226 observations** across **457 tickers**; the ∼4,600 dropped observations reflect incomplete price windows in 2016–2017 and late 2025.

The tone-scoring dataset covers **478 firms**, while the reduction to 457 firms in the regression sample is due to missing price data or incomplete return windows for a subset of firms. Descriptive statistics are in Table A2 (Appendix).

3 Methodology

Preprocessing and chunking. Raw transcripts are normalized, segmented by speaker (management / analyst / operator), and split into semantic chunks of at most 510 tokens. No stemming or stopword removal is applied, as negations carry critical semantic content for hedging detection. The full pipeline produces **952,520 chunks**; restricting to `speaker_type = 'management'` and `section = 'prepared_remarks'` yields the **183,491 chunks** used for scoring.

Knowledge-distillation labeling. Training data combine the *financial_phrasebank* dataset (Malo et al., 2014) (3,876 expert-annotated financial sentences) with 939 LLM-generated labels from a stratified sample of earnings-call chunks produced via the Groq API (Llama-3.1-8B), for a total of **4,120 training examples**. Each chunk is assigned one of three labels — *cautious*, *optimistic*, or *negative* — with cautious labels dominating at 53.6%, reflecting the hedging-heavy register of prepared remarks (Table A3, Appendix).

FinBERT fine-tuning. The student model is built on `ProsusAI/finbert` (Araci, 2019), a BERT architecture pre-trained on financial text. The classification head maps the 768-dimensional backbone to three classes via softmax and cross-entropy loss. The labeled dataset is split 80/10/10 (stratified by sector and year); fine-tuning uses AdamW (lr $= 2 \times 10^{-5}$, batch size 16, 3 epochs) on a T4 GPU.

The key regression variable is:

$$\texttt{cautious_share}_{it} = \text{mean}\{P(\texttt{cautious})_k : k \in \text{prepared-remarks chunks of call } (i,t)\}$$

aggregated to **15,039 call-level observations**. Higher values indicate calls dominated by hedging and defensive language.

Econometric specification. We estimate two OLS panel models with HC3-robust standard errors and sector, quarter, and crisis fixed effects. The *baseline* uses only financial controls:

$$\texttt{rv_post}_{it} = \alpha + \beta_1\,\texttt{rv_hist}_{it} + \beta_2\,\texttt{vix}_t + \beta_3\,\texttt{covid}_t + \beta_4\,\texttt{rate_hike}_t + \boldsymbol{\gamma}'\mathbf{d}_i^{\text{sector}} + \boldsymbol{\delta}'\mathbf{d}_t^{\text{quarter}} + \varepsilon_{it} \quad (1)$$

The *full* model augments it with the tone variable:

$$\begin{aligned}\texttt{rv_post}_{it} = \alpha &+ \beta_1\,\texttt{rv_hist}_{it} + \beta_2\,\texttt{vix}_t + \beta_3\,\texttt{cautious_share}_{it} \\ &+ \beta_4\,\texttt{covid}_t + \beta_5\,\texttt{rate_hike}_t + \boldsymbol{\gamma}'\mathbf{d}_i^{\text{sector}} + \boldsymbol{\delta}'\mathbf{d}_t^{\text{quarter}} + \varepsilon_{it}\end{aligned} \quad (2)$$

where `covid` = 1 for calls between January–September 2020, and `rate_hike` = 1 for calls between January 2022 and June 2023. Sector fixed effects absorb cross-sectional heterogeneity in baseline volatility; quarter fixed effects capture seasonal and macro patterns. Realized volatility starts strictly after the call date (t+1), ensuring no look-ahead bias. VIFs confirm the absence of multicollinearity (max VIF = 1.33; see Table A6, Appendix).

4 Results

FinBERT classification performance. The fine-tuned model achieves an overall accuracy of **86%** on the held-out test set (412 chunks). Macro-average F1 is 0.86, with per-class F1 of 0.81 (optimistic), 0.88 (cautious), and 0.89 (negative). Full metrics are in Table A4 (Appendix).

Regression results. Table 1 reports OLS estimates. In the baseline, `rv_hist`, `vix`, and the crisis dummies are all highly significant ($p < 0.001$), explaining $R^2 = 0.445$. Adding `cautious_share` leaves the fit entirely unchanged.

Table 1: OLS estimates: post-call realized volatility ($N = 10{,}226$)

	Baseline (1)	**Full model (2)**
`cautious_share`		0.0007
		(0.0102)
`rv_hist`	0.4266***	0.4265***
	(0.0191)	(0.0192)
`vix`	0.0040***	0.0040***
	(0.0005)	(0.0005)
`covid`	0.0815***	0.0815***
	(0.0129)	(0.0130)
`rate_hike`	0.0259***	0.0259***
	(0.0064)	(0.0064)
Sector FE	Yes	Yes
Quarter FE	Yes	Yes
Crisis dummies	Yes	Yes
R^2	0.4447	0.4447
ΔR^2		0.000
N	10,226	10,226

Notes: Dependent variable: `rv_post` (annualized realized volatility, 20 trading days post-call). HC3-robust standard errors in parentheses. `covid` = 1 for January–September 2020; `rate_hike` = 1 for January 2022–June 2023. $^{***}p < 0.001$. 95% CI for $\hat{\beta}_3$: $[-0.019, 0.021]$; $p = 0.945$. VIF max = 1.33.

Economic magnitude. A 10 percentage-point increase in `cautious_share` — moving from the median call (38% cautious chunks) to the 75th percentile (49% cautious chunks) — is asso-

ciated with an increase of only 0.007 annualized percentage points in post-call realized volatility, an effect indistinguishable from zero ($p = 0.945$; 95% CI: $[-0.019, 0.021]$). By contrast, both crisis dummies are highly significant: the COVID-19 shock raises post-call volatility by 8.2 percentage points ($p < 0.001$), and the rate-hiking cycle by 2.6 percentage points ($p < 0.001$). The improvement in R^2 from 0.437 to 0.445 is entirely attributable to these crisis controls, not to the tone variable.

5 Discussion and Conclusion

The estimated coefficient on `cautious_share` is statistically indistinguishable from zero and the full model R^2 is identical to the augmented baseline at 44.5%. This null result is informative rather than a failure. Large S&P 500 firms operate under intensive analyst scrutiny; cautious language in *prepared remarks* may be efficiently priced by market participants, leaving no detectable residual signal. Several well-designed studies report modest or conditional tone effects (Price et al., 2012; Huang et al., 2014), and it is plausible that such measures are subsumed by the VIX and historical volatility once sector and quarter fixed effects are included.

Limitations. *First*, knowledge-distillation labels may introduce systematic noise for ambiguous management language combining optimistic guidance with cautious qualifications; the negative class (9.4% of training data) may be under-represented for reliable generalization. *Second*, the absence of an earnings surprise control (actual EPS vs. analyst consensus) is an important omission: calls with large negative surprises may simultaneously elicit more cautious tone *and* higher post-call volatility, creating a confound. *Third*, the S&P 500 restriction introduces survivorship bias and limits generalizability to smaller firms where information environments are less efficient and tone signals may carry more incremental value.

Future extensions. The most immediate improvement is recovering the ∼4,600 dropped observations from 2016–2017 and late 2025. Beyond data coverage, incorporating earnings surprise controls, decomposing tone between prepared remarks and Q&A, and testing regime-conditional effects via interaction terms between `cautious_share` and the crisis dummies are natural next steps. More broadly, this paper illustrates the potential of domain-adapted NLP pipelines to extract risk-relevant signals from unstructured corporate communication, and provides a reproducible foundation for these extensions.

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Appendix

Table A1: Pipeline stages and dataset sizes

Stage	Count
Raw scraping (pre-deduplication)	114,466
Raw corpus (post-deduplication)	113,990
S&P 500 filtered transcripts	14,963
Total chunks (full pipeline)	952,520
Prepared-remarks chunks (main analysis)	183,491
LLM distillation labels	4,120
Call-level tone scores	15,039
Market / volatility panel	14,847
Regression sample (merged features)	10,226

Table A2: Descriptive statistics — financial panel ($N = 14{,}847$)

Variable	Mean	Std.	Min	Max	Interpretation
`rv_post`	0.274	0.181	0.049	3.629	Annualized vol., 20 days post-call
`rv_hist`	0.293	0.165	0.080	2.368	Annualized vol., 60 days pre-call
`vix`	18.990	7.042	9.140	76.450	CBOE VIX on call date

Source: Yahoo Finance / yfinance. The mild decline from `rv_hist` (0.293) to `rv_post` (0.274) is consistent with a partial resolution of uncertainty following the disclosure event. Maximum values of 3.629 and 76.45 are attributable to the COVID-19 market stress of early 2020.

Table A3: Label distribution — knowledge-distillation dataset (4,120 chunks)

Label	Count	Share (%)
Optimistic	1,522	36.9
Cautious	2,210	53.6
Negative	388	9.4
Total	**4,120**	100.0

Table A4: FinBERT classification performance — held-out test set (412 chunks)

Class	Precision	Recall	F1-score	Support
Optimistic	0.87	0.76	0.81	152
Cautious	0.85	0.91	0.88	221
Negative	0.86	0.92	0.89	39
Accuracy			0.86	412
Macro avg	0.86	0.87	0.86	412

Note: Metrics computed on the stratified 10% held-out split. Fine-tuning uses AdamW (lr $= 2 \times 10^{-5}$*, batch size 16, 3 epochs) on* `ProsusAI/finbert`*.*

Table A5: Descriptive statistics — call-level tone scores (15,039 calls, 478 tickers)

Variable	Mean	Std.	Min	P25	Med.	P75	Max
`cautious_share`	0.399	0.141	0.042	0.294	0.383	0.488	0.970
`optimistic_share`	0.583	0.145	0.013	0.492	0.600	0.691	0.949
`negative_share`	0.018	0.007	0.009	0.014	0.016	0.019	0.235

Scores are mean-pooled across all prepared-remarks chunks of each call. The three shares sum to one by construction. The very low `negative_share` *(mean: 0.018) reflects the rarity of explicitly negative language in scripted institutional communication.*

Table A6: Variance Inflation Factors — full model regressors

Variable	VIF
`cautious_share`	1.045
`rv_hist`	1.331
`vix`	1.288

All VIFs well below the conventional threshold of 5, confirming the absence of multicollinearity among regressors.